Oceanic Mixed Bottom Layers

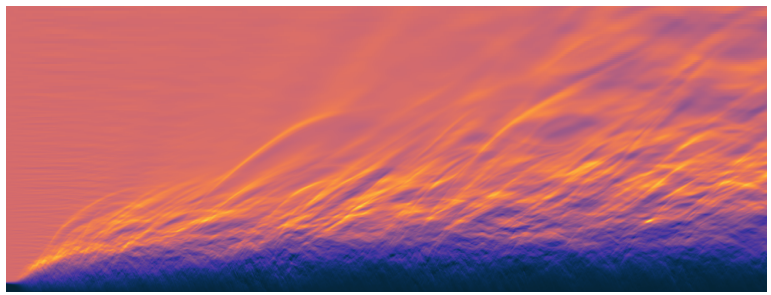
Summer Research Festival 2026

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Monday 12th October 2026



Overview

1. Background
2. Ekman model
3. Stokes model
4. Simulating BML
5. Predicting BML evolution
6. Summary

What is are oceanic bottom mixed layers?

- Form in the deep, abyssal ocean
- Exhibits near-uniform water temperature and salinity (as opposed to the stratification/layering of the interior ocean).
- Mixed layer height typically $10 \sim 100\,\mathrm{m}$

Why is it of interest?

- Climate control
- Marine ecosystems

Background

Observations in the abyssal ocean show well-mixed layers near the seafloor.

- The Hatteras Abyssal Plain (in the Atlantic Ocean) exhibits these bottom mixed layers in the abyssal ocean.
- Observed layer depth scales with mean current speed, suppressed by background stratification. [Armi and Millard Jr 1976]
- Global surveys show especially thick bottom mixed layers in other weakly stratified regions. [Huang et al. 2019]

Hatteras Abyssal Plain

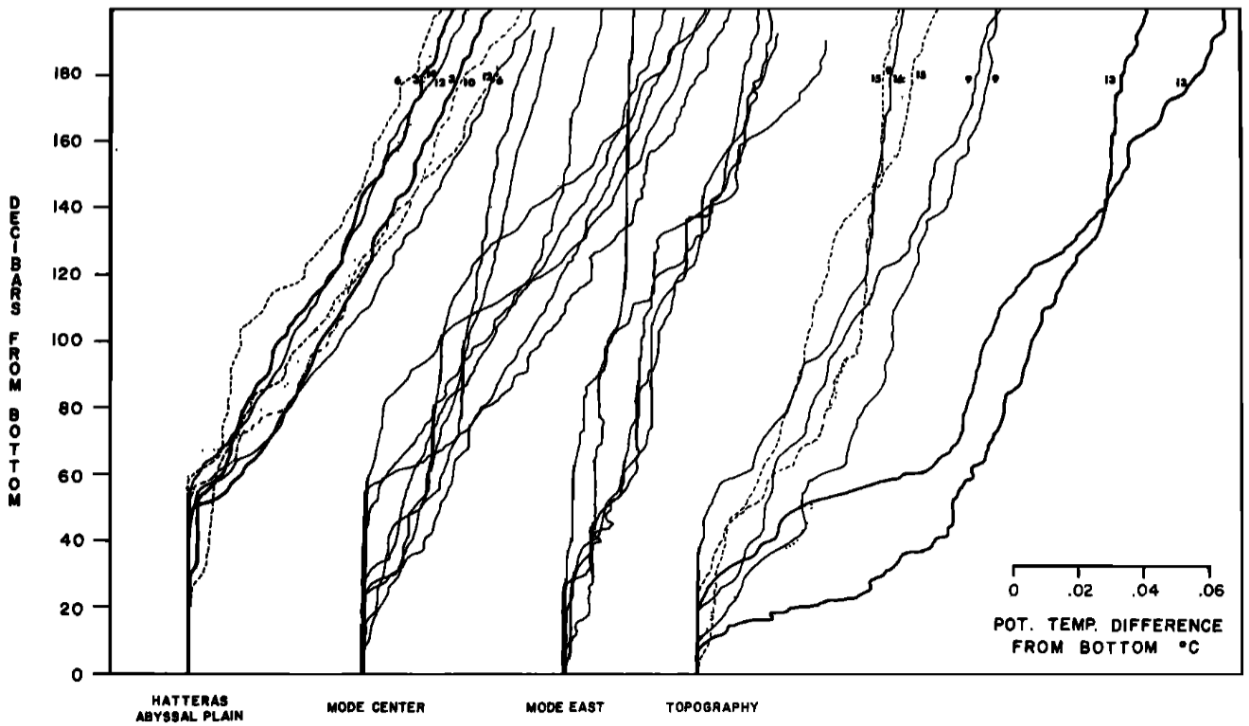


Figure: Potential temperature profiles relative to ocean bottom at various locations around Hatteras Abyssal Plain. [Armi and Millard Jr 1976]

Causes of thick BML

In this project, we explored the following question:

What physical mechanism produces these thick bottom mixed layers (BML)?

As there is a mixing process which generates turbulence, we investigated two different sources of energy:

- *Benthic storms*: episodic bursts of intense bottom currents, with typical speeds $\sim 20\,\mathrm{cm\,s^{-1}}$ over weeks
  - $\rightarrow$ Ekman model
- *Tides*: periodic tidal current with typical speeds on the order of $1\,\mathrm{cm\,s^{-1}}$ to $2\,\mathrm{cm\,s^{-1}}$, and wave breaking of internal tides
  - $\rightarrow$ Tidal/Stokes model

Approximations & assumptions

We will make use of the following approximations throughout:

- *Boussinesq approximation*: neglect density variations $\rho' \ll \rho_0$, but retained only in the buoyancy term of the momentum equation
- *Incompressibility*: assume we have incompressible flow

$$\nabla \cdot \mathbf{u} = 0$$

- *Mass conservation/continuity equation*:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad \Longleftrightarrow \quad \frac{D\rho}{Dt} = 0$$

Ekman layer near rigid boundary

Assume steady far-stream flow $U_\infty \hat{\mathbf{x}}$ in geostrophic balance far from rigid boundary at $z = 0$:

$$fU_\infty = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

We will use an f-plane approximation, where $f = 2\Omega \sin\theta$ is constant.
Ekman layers form due to balance of *Coriolis* and *friction* forces.

Analytic solution for Ekman layer

Look for steady-state solution:

$$-fv = \nu \frac{\mathrm{d}^2 u}{\mathrm{d}z^2}, \qquad +fu = \nu \frac{\mathrm{d}^2 v}{\mathrm{d}z^2} + fU_\infty$$

Boundary conditions:

- *No-slip*: $\mathbf{u} = 0$ at $z = 0$
- *Far-field*: $\mathbf{u} \to U_\infty \hat{\mathbf{x}}$ as $z \to +\infty$.

Rewrite $V = u + iv$, and find analytic solution

$$V = U_\infty \left[1 - \exp\left(-\frac{(1+i)z}{\delta_E} \right) \right]$$

where $\delta_E = \sqrt{2\nu/f}$.

Analytic solution for Ekman layer

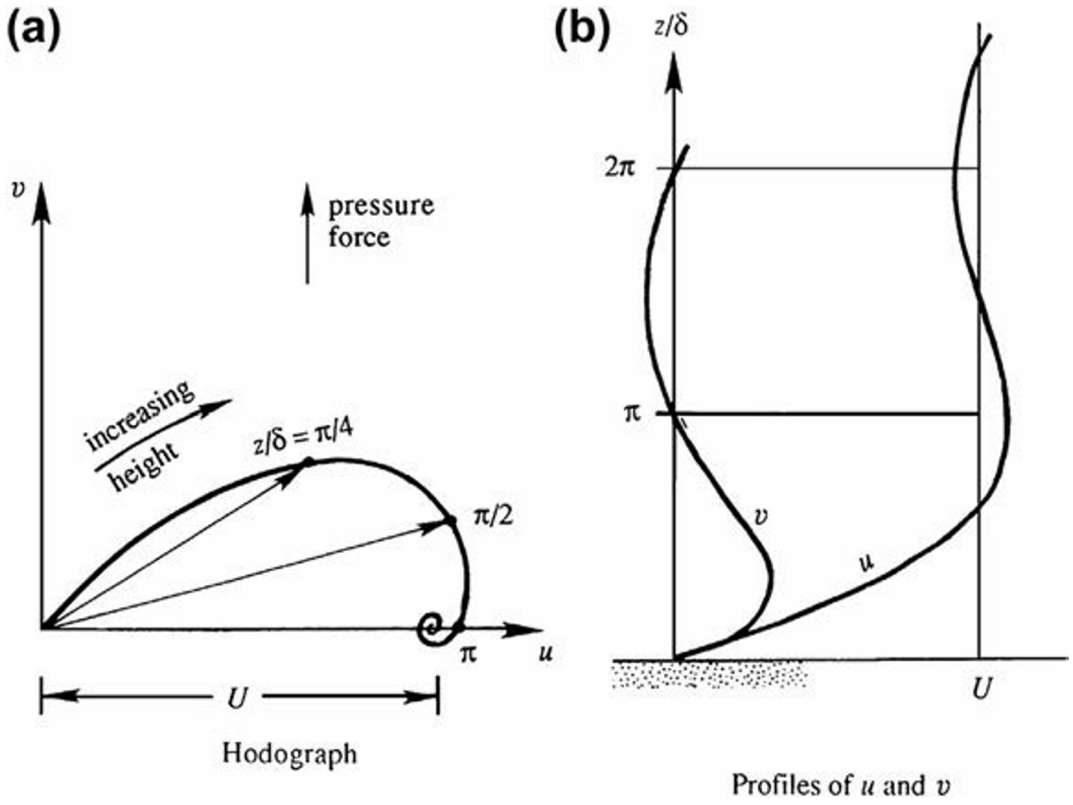


Figure: Hodograph, and plot of velocity components with depth for the analytic Ekman layer solution along a rigid boundary. [Kundu, Cohen, and Dowling 2015]

Stokes boundary layer with a stationary flat plate

We have a stationary, rigid boundary at $z = 0$, and the fluid far away has velocity $U_0 \hat{\mathbf{x}} \cos \omega t$.

Seek a solution of the form $\mathbf{u} = (u(z,t), 0, 0)$:

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial^2 z}$$

Boundary conditions:

- *No-slip*: $\mathbf{u} = 0$ at $z = 0$
- *Outer flow*: $\mathbf{u} \to U_0 \hat{\mathbf{x}} \cos \omega t$ as $z \to \infty$

Analytic solution for Stokes layer (stationary plate)

The harmonic solution is given by:

$$u = U_0 \left[\cos(\omega t) - e^{-z/\delta_S} \cos\left(\omega t - \frac{z}{\delta_S} \right) \right],$$

where $\delta_S = \sqrt{2\nu/\omega}$. [see IB Fluid Dynamics ES2]

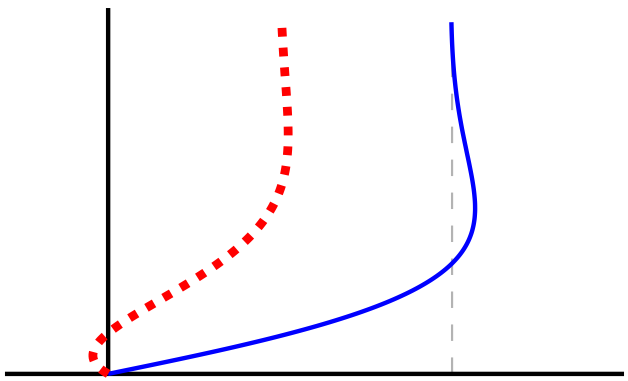


Figure: Velocity profile for a stationary flat plate with an oscillating far-field flow.

Simulating and implementating both models

- Used *Oceananigans* in Julia
- Domain:
  - Periodic in x and y, and vertical height $\sim 100\,\mathrm{m}$
  - Time evolution ~ 1 week
- Stretched grid: finer spacing towards seafloor
- Closure: Anisotropic minimum dissipation (AMD) scheme
- Advection: Centred fourth-order scheme
- Sponge layer above the domain
- Quadratic drag boundary condition at $z = 0$, using drag coefficient

$$c_D = \left[\kappa \ln\left(\frac{z_1}{z_0}\right)\right]^2$$

Parameters

To compare the models, we control the following parameters

Background velocity	$U_0 = U_\infty = 0.04\,\mathrm{m\,s^{-1}}$
Frequency	$\omega = f = 10^{-4}\,\mathrm{s^{-1}}$
Prandtl number	$\mathrm{Pr} = \nu_0/\kappa_0 = 10$
Molecular kinematic viscosity	$\nu_0 = 10^{-6}\,\mathrm{m^2\,s^{-1}}$
Molecular diffusivity	$\kappa_0 = 10^{-7}\,\mathrm{m^2\,s^{-1}}$
Roughness length	$z_0 = 1.6 \times 10^{-3}\,\mathrm{m}$

We vary the stratification parameter $r = N/f = N/\omega$, where N is the Brunt-Väisälä/buoyancy frequency far from the mixed layer.

Setup for Ekman model

$$\frac{Du}{Dt} \boxed{-fv} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xz}}{\partial z}$$
$$\frac{Dv}{Dt} \boxed{+fu} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \boxed{+fU_\infty}$$

- Steady far-stream velocity $\boxed{U_\infty \hat{\mathbf{x}}}$
- Includes Coriolis effects

Setup for Stokes model

$$\frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xz}}{\partial z} + U_0 \omega \cos \omega t$$
$$\frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{yz}}{\partial z}$$

- Oscillatory far-stream velocity $U_0 \hat{\mathbf{x}} \sin \omega t$
- No Coriolis effects

Replicating results (Ekman)

Initial demonstrations show simulations behave as expected, by comparing with Taylor and Sarkar 2008 and Gayen, Sarkar, and Taylor 2010.

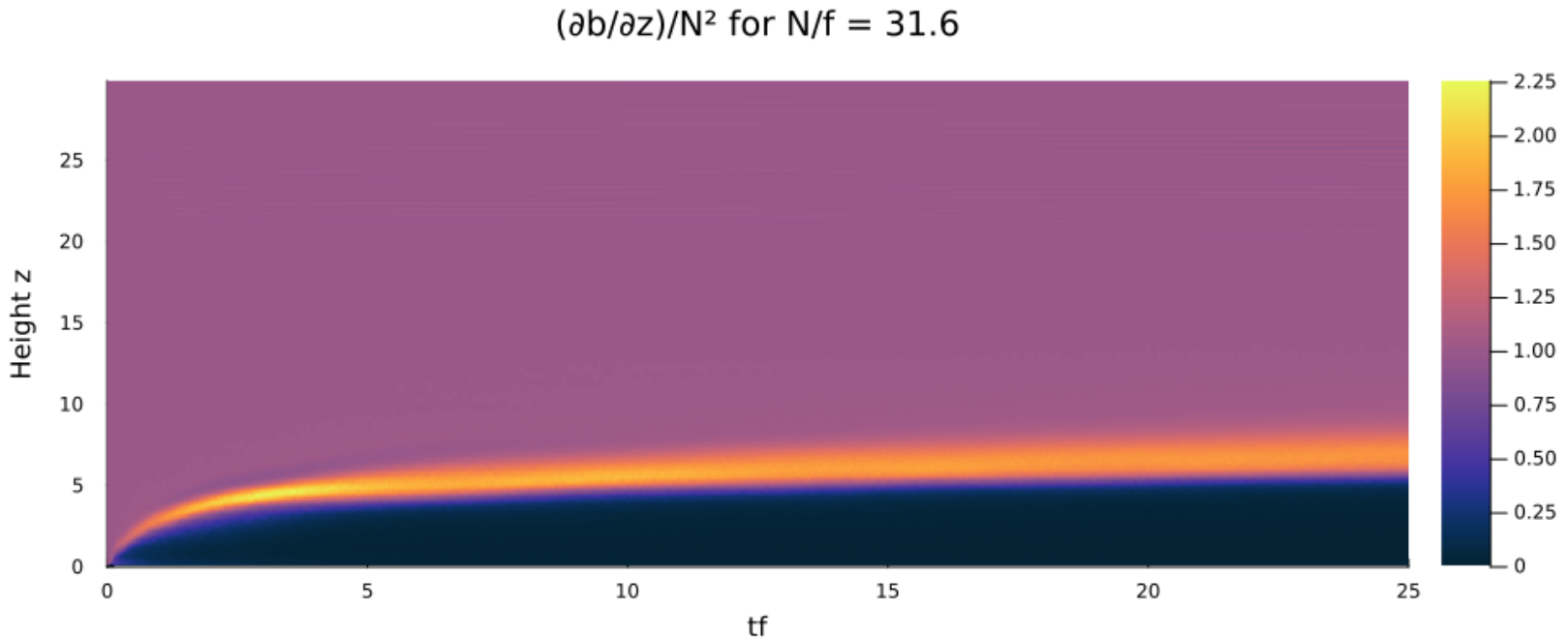


Figure: Buoyancy gradient plot for Ekman model, initialised with a linear stratification profile, which is consistent with Taylor and Sarkar 2008.

Replicating results (Ekman)

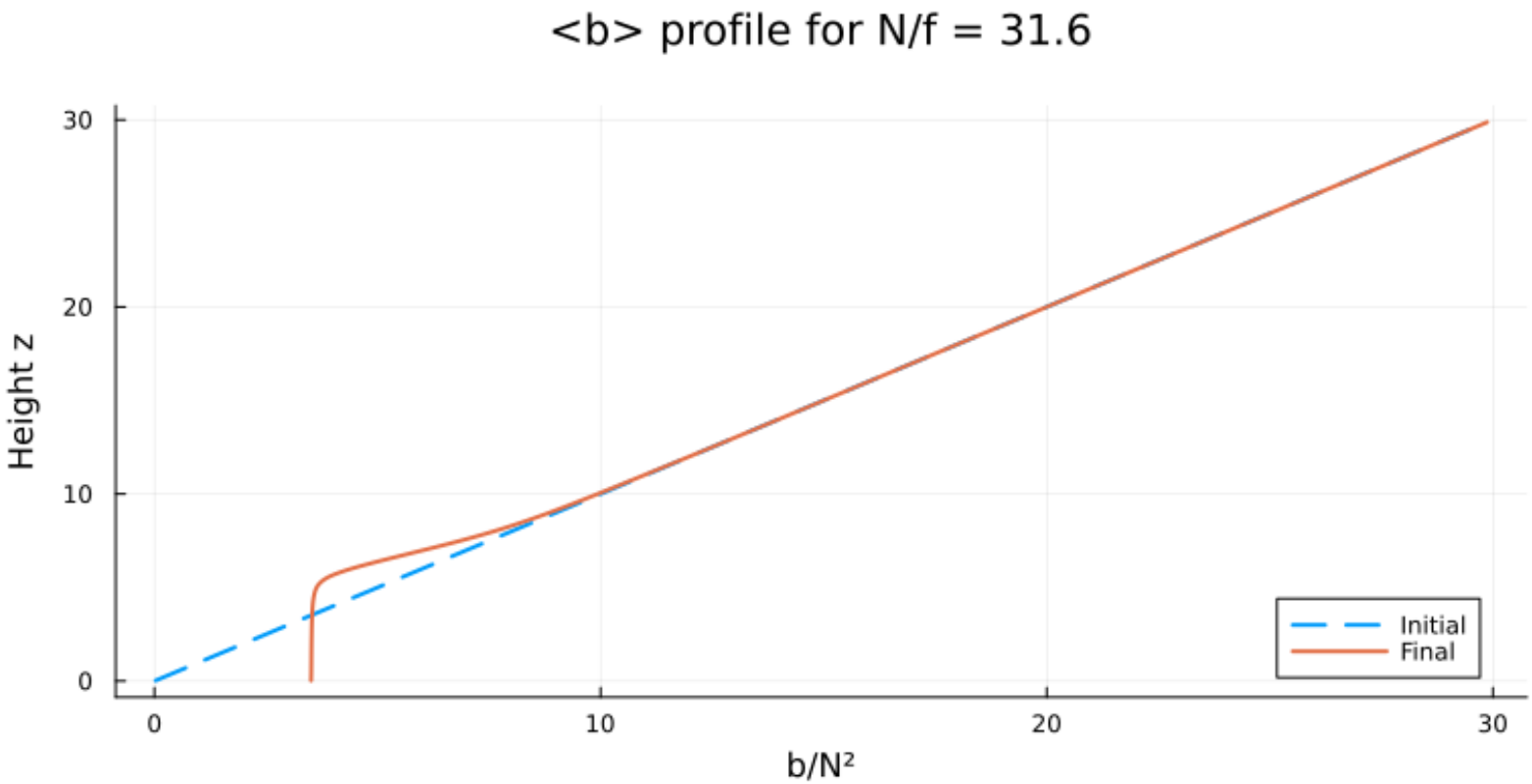


Figure: Buoyancy profile for Ekman model, as before, which is consistent with Taylor and Sarkar 2008.

Replicating results (Stokes)

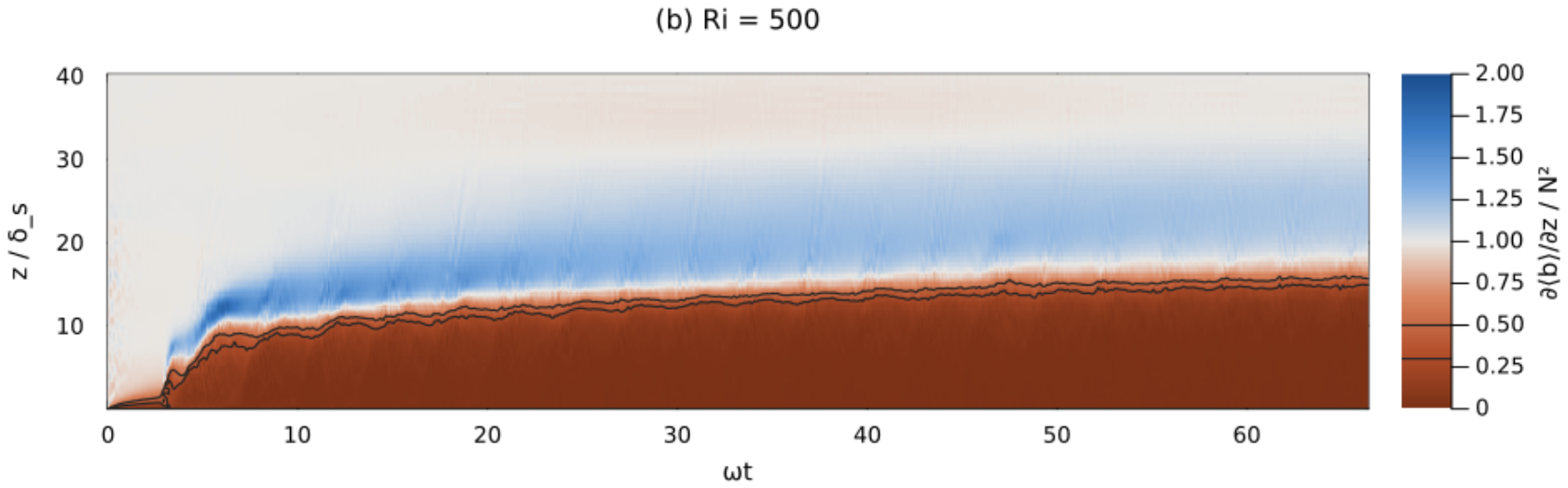


Figure: Buoyancy gradient plot for Stokes model, initialised with a linear stratification profile, which is consistent with Gayen, Sarkar, and Taylor 2010.

Replicating results (Stokes)

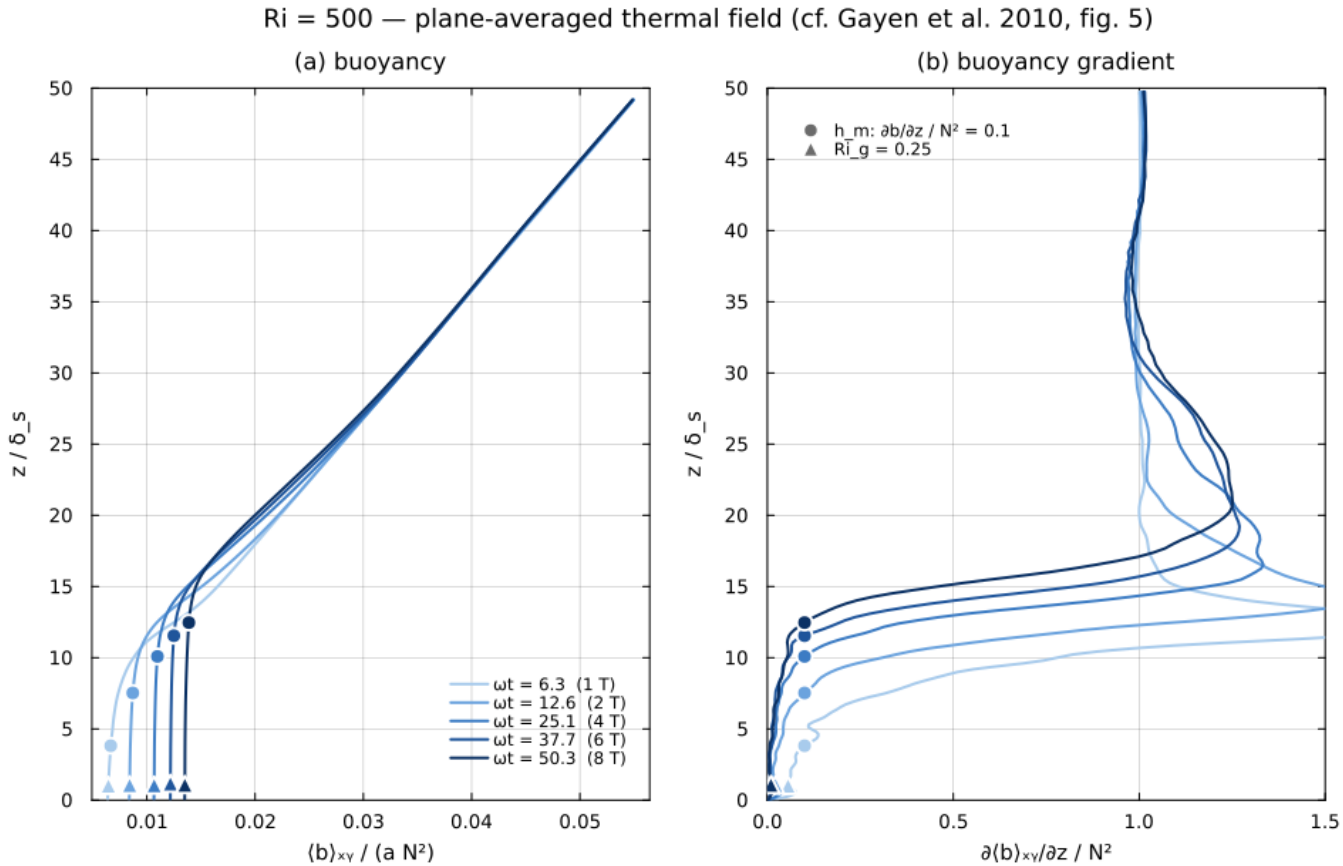


Figure: Buoyancy profiles at different times for Stokes model, as before, which is consistent with Gayen, Sarkar, and Taylor 2010.

Animating our fluid

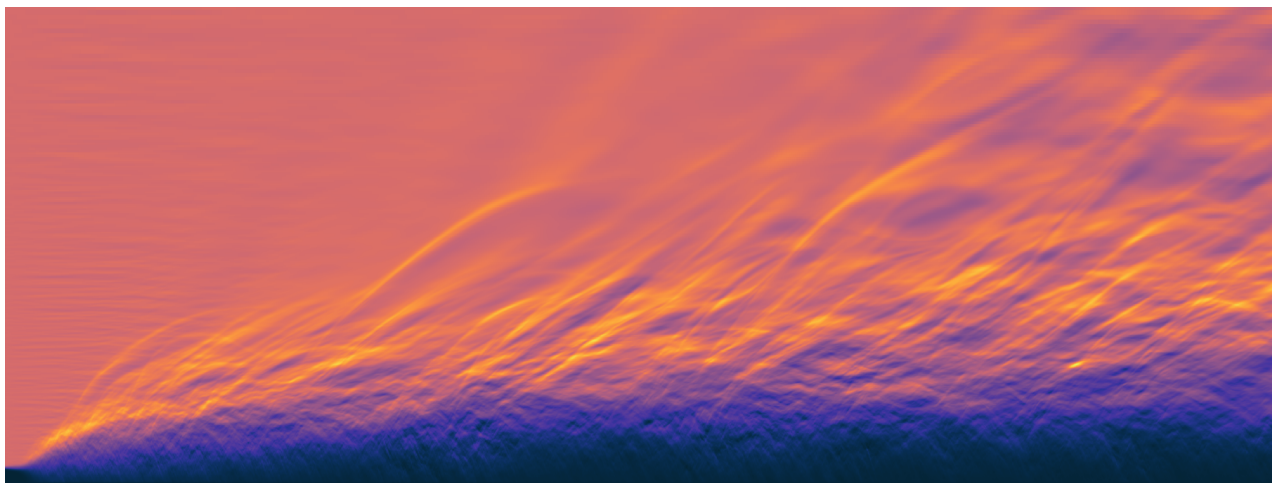


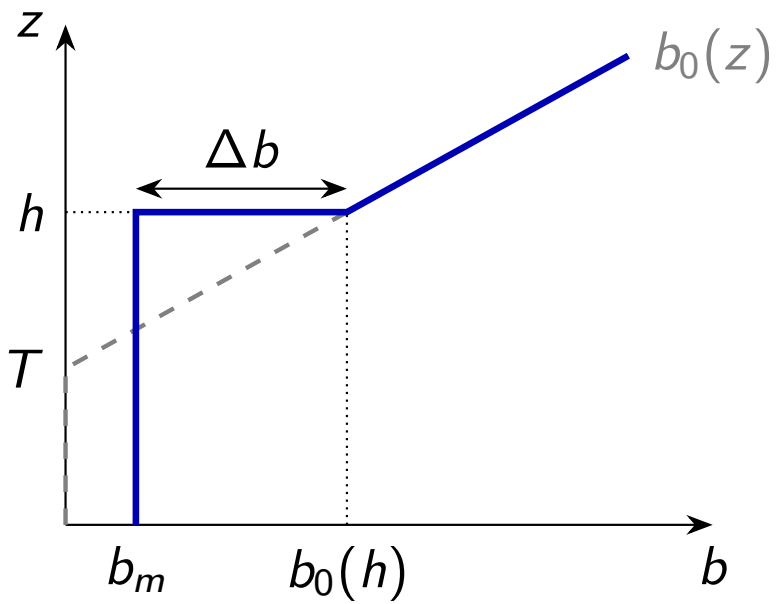
Figure: Ekman flow animation

Initial observations

- In all simulations, mixed layer height did not reach near the observed $\sim 60\,\mathrm{m}$.
- However, we observe two phases of growth:
  - Fast, short initial growth (for a time period of $\sim 2tf = 2\omega t$)
  - Slow, long-term growth
- Instead, turned to theory to extract a timescale from numerical simulation
  - Focused on turbulent kinetic energy and turbulent diffusivity

Time evolution of the mixed layer height

We derive an evolution equation of the mixed layer height, $h(t)$. For a well-mixed layer, the buoyancy profile can be approximated by a piecewise linear profile:



Time evolution of the mixed layer height

Using conservation of buoyancy and the buoyancy flux relations

$$\frac{\partial \tilde{b}}{\partial t} + \frac{\partial F_b}{\partial z} = 0 \qquad F_b + K_T \frac{\partial \tilde{b}}{\partial z} = 0,$$

it can be shown that the evolution of the mixed layer height is given by

$$\frac{dh}{dt} = \frac{2K_T(h)}{h},$$

where K_T is the turbulent diffusivity.

Finding $K_T(h)$

We want to find $K_T(h)$ at mixed layer height h, to extract a timescale for mixed layer growth from the evolution equation.

- Find the time- and horizontally-averaged turbulent kinetic energy (TKE), where

$$\mathrm{TKE} = \frac{1}{2}|\mathbf{u}'|^2 = \frac{1}{2}|\mathbf{u} - \langle\tilde{\mathbf{u}}\rangle|^2.$$

- Find, using data fitting, $\mathrm{TKE}(h)$, $\mathrm{Ri}(h)$, as well as the dimensionless quantity $K_T^*(\mathrm{TKE}, \mathrm{Ri})$, where

$$K_T^* = \frac{NK_T}{\mathrm{TKE}}.$$

$\Longrightarrow$ Find $K_T(h)$ with the above.

Finding $K_T(h)$

Define the length scale

$$L_K \sim \frac{K_T}{\sqrt{\text{TKE}}},$$

which is the typical size of eddies.
Our two candidate length scales are

$$L_N = \frac{\sqrt{\text{TKE}}}{N}, \qquad L_S = \frac{\sqrt{\text{TKE}}}{S}.$$

where $S = \langle |\partial \tilde{\mathbf{u}}/\partial z| \rangle$ is the time- and horizontally-averaged shear at $z = h$. We found that L_N was the relevant length scale at greater stratifications, whilst L_S is the relevant at lower stratifications.

Finding $K_T(h)$

We aim to fit K_T, which incorporates both length scales.
We now look at

$$\frac{L_S}{L_N} = \sqrt{\mathrm{Ri}}, \quad \mathrm{Ri} = \frac{N^2}{S^2},$$

where Ri is the gradient Richardson number. We can then fit

$$\frac{1}{L_K} \propto \frac{1}{L_S} + \frac{1}{L_N},$$

which can be rewritten in the non-dimensionalised form

$$K_T^* \propto \frac{\sqrt{\mathrm{Ri}}}{1 + \sqrt{\mathrm{Ri}}},$$

where $K_T^* = NK_T/\mathrm{TKE}$.

Finding $K_T(h)$

Finally, we can fit $\mathrm{Ri} = \mathrm{Ri}(h)$, so that we can find $K_T(h)$:

$$\mathrm{Ri}(h) \longrightarrow K_T^*(\mathrm{TKE}, \mathrm{Ri}(h)) \longrightarrow K_T(h)$$

$$\uparrow$$

$$\mathrm{TKE}(h)$$

Fitting Stokes results

In the Stokes case, we can fit our data, similar to an approach in Weatherly and Martin 1978:

$$K_T^* = 0.412 \, \frac{\sqrt{\mathrm{Ri}}}{1 + \sqrt{\mathrm{Ri}}}$$

$$\mathrm{TKE} = 3.43 \, u_*^2 \exp\left(\frac{-1.47 \, z}{\delta}\right)$$

$$S(h) = 6.20 \times 10^{-3} \, e^{-h/3.2}$$

with $\delta = 0.4 \, (u_*/\omega) \, (1 + N^2/\omega^2)^{-0.020}$ and $u_* = 1.1 \times 10^{-3} \, \mathrm{m\,s^{-1}}$.

Finding $K_T(h)$

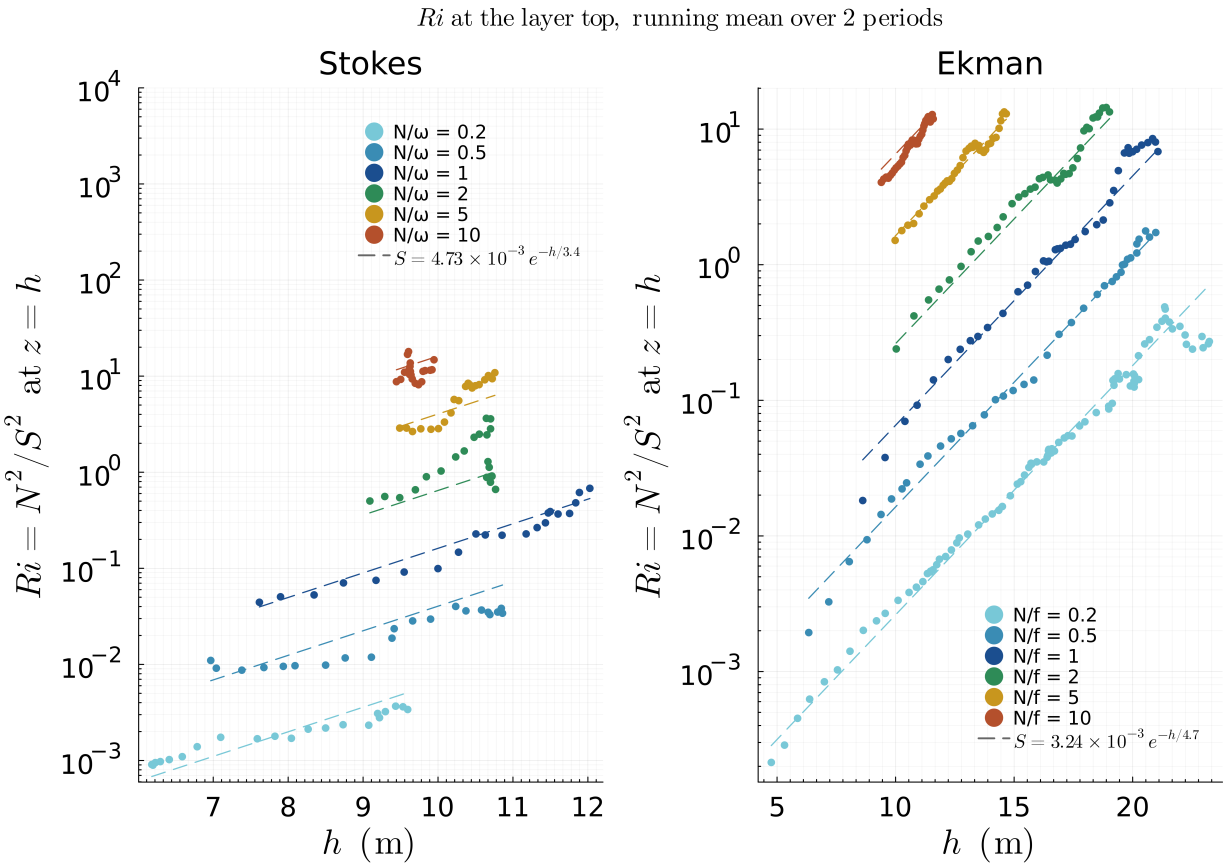


Figure: Plot of Richardson number against layer height for both models, with line of best fit for different stratifications.

Fitting Stokes results

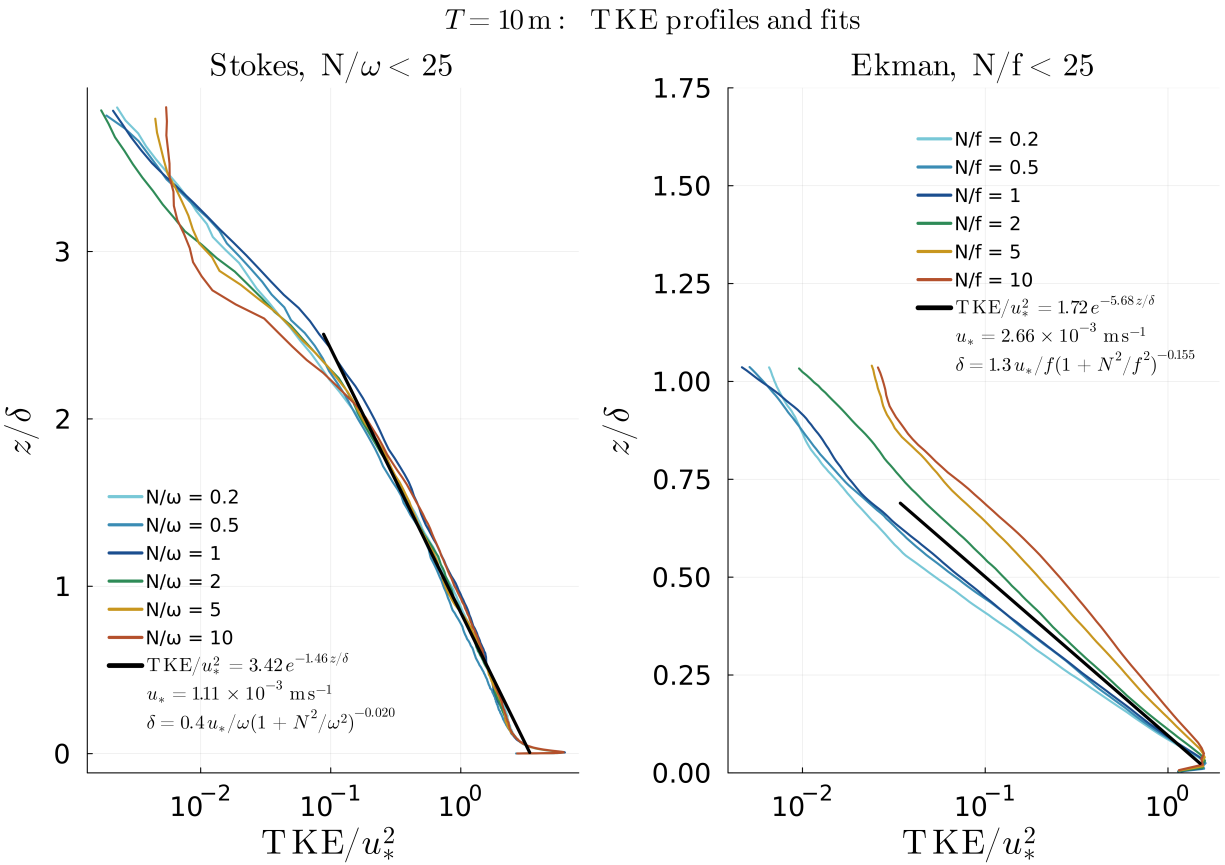


Figure: TKE profiles at different stratifications for both models, as well as a linear line of best fit the near seafloor.

Fitting Stokes results

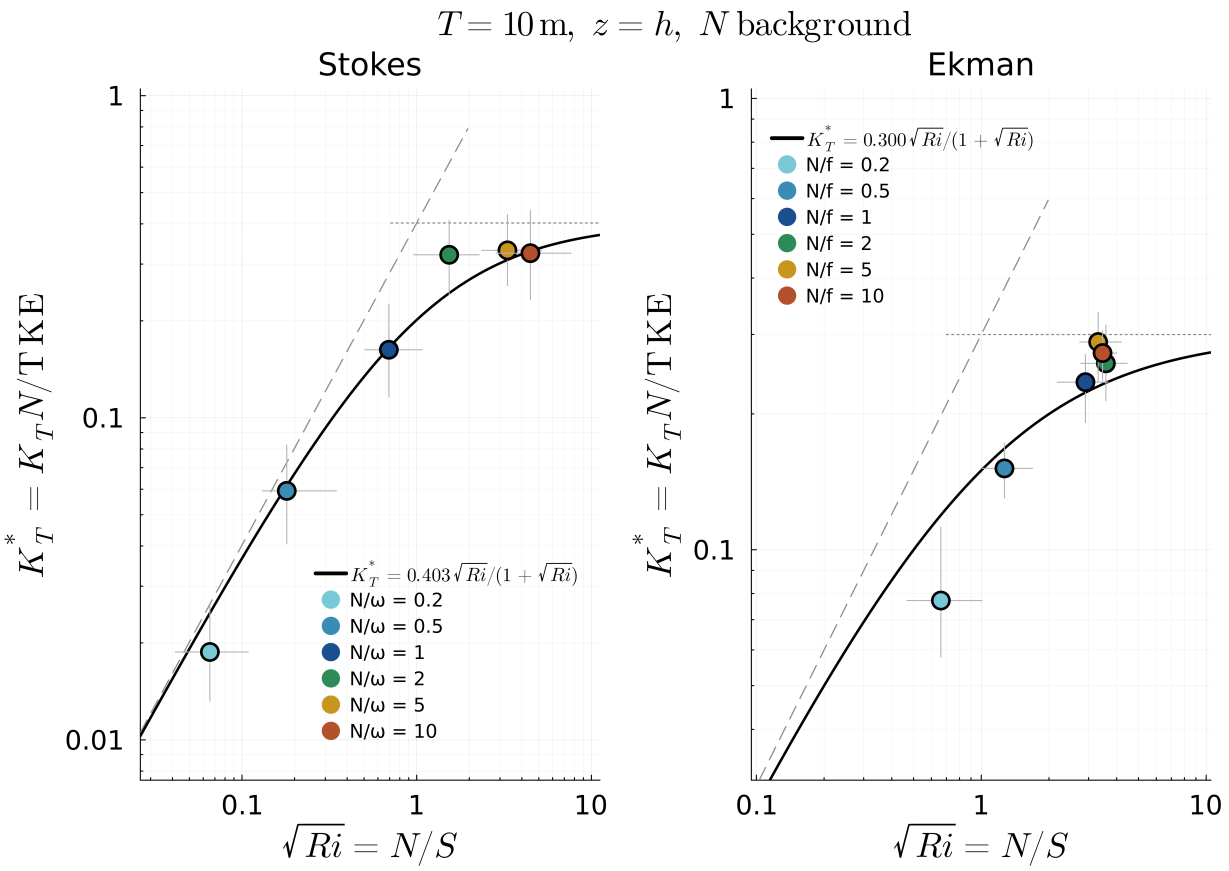


Figure: Plot of K_T^* against $\sqrt{\mathrm{Ri}}$ with different stratifications for both models, along with a fitted curve.

Fitting Stokes results

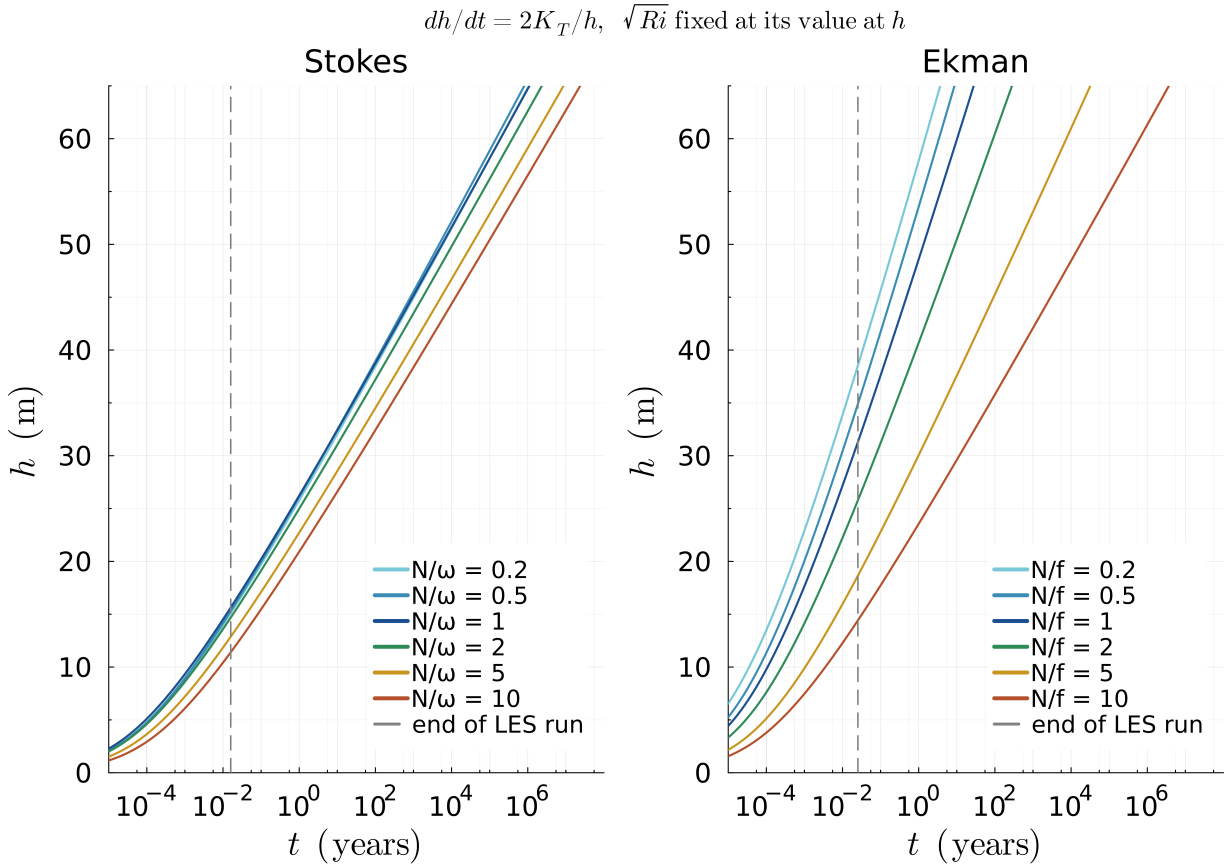


Figure: Plot of h over time numerically for both models with our evolution equation.

Characteristic timescales for growth (Stokes)

N/ω	$h = 10\,\mathrm{m}$	$h = 20\,\mathrm{m}$	$h = 30\,\mathrm{m}$	$h = 40\,\mathrm{m}$	$h = 50\,\mathrm{m}$	$h = 60\,\mathrm{m}$
0.2	1.7×10^{-3}	1.1×10^{-1}	4.9×10^{0}	1.8×10^{2}	6.3×10^{3}	2.1×10^{5}
0.5	1.6×10^{-3}	1.0×10^{-1}	4.2×10^{0}	1.5×10^{2}	5.0×10^{3}	1.6×10^{5}
1	1.4×10^{-3}	9.2×10^{-2}	4.2×10^{0}	1.7×10^{2}	6.0×10^{3}	2.1×10^{5}
2	1.9×10^{-3}	1.5×10^{-1}	7.0×10^{0}	3.0×10^{2}	1.1×10^{4}	4.3×10^{5}
5	4.1×10^{-3}	3.3×10^{-1}	1.8×10^{1}	8.4×10^{2}	3.6×10^{4}	1.5×10^{6}
10	8.1×10^{-3}	7.0×10^{-1}	4.0×10^{1}	1.9×10^{3}	9.1×10^{4}	4.0×10^{6}

Table: Characteristic timescales (in years) for the mixed layer to reach certain heights with the Stokes model at different stratifications.

Characteristic timescales for growth (Ekman)

We repeat the same analysis for the Ekman model.

N/f	$h = 10\,\mathrm{m}$	$h = 20\,\mathrm{m}$	$h = 30\,\mathrm{m}$	$h = 40\,\mathrm{m}$	$h = 50\,\mathrm{m}$	$h = 60\,\mathrm{m}$
0.2	3.2×10^{-5}	4.7×10^{-4}	4.1×10^{-3}	3.1×10^{-2}	2.1×10^{-1}	1.3×10^{0}
0.5	5.9×10^{-5}	9.0×10^{-4}	8.3×10^{-3}	6.4×10^{-2}	4.6×10^{-1}	3.1×10^{0}
1	9.8×10^{-5}	1.7×10^{-3}	1.7×10^{-2}	1.5×10^{-1}	1.2×10^{0}	9.3×10^{0}
2	2.3×10^{-4}	5.0×10^{-3}	6.7×10^{-2}	7.7×10^{-1}	8.1×10^{0}	8.2×10^{1}
5	9.5×10^{-4}	3.6×10^{-2}	8.9×10^{-1}	1.9×10^{1}	3.7×10^{2}	6.8×10^{3}
10	3.2×10^{-3}	2.3×10^{-1}	1.1×10^{1}	4.3×10^{2}	1.6×10^{4}	5.7×10^{5}

Table: Characteristic timescales (in years) for the mixed layer to reach certain heights with the Ekman model at different stratifications.

Summary and future discussion

- Looked at Stokes and Ekman model for formation of oceanic BML
- Found an evolution equation for the mixed layer height, which can be autonomously evolved in time, utilising data fitting.
- In both cases, the timescales derived for layer height to reach $\sim 60\,\mathrm{m}$ was $\sim 10^5\,\mathrm{yr}$ for Stokes, and only a few years for the Ekman model, in the regime of the abyssal ocean.

- Why is there significantly faster layer formation for the Ekman than Stokes model?
- Horizontal advection: can change stratification non-uniformly with height

Thank you!

Many thanks to our supervisors Prof. John Taylor, Dr Bethan Wynne-Cattanach, and Dr Yuchen Ma, and all the other wonderful people we have been grateful to meet on SRIM! 😄

We would also like to thank the EPSRC for providing funding for our summer project.



References

- Armi, Laurence and Robert C Millard Jr (1976). "The bottom boundary layer of the deep ocean". In: *Journal of Geophysical Research* 81.27, pp. 4983–4990.
- Gayen, Bishakhdatta, Sutanu Sarkar, and John R Taylor (2010). "Large eddy simulation of a stratified boundary layer under an oscillatory current". In: *Journal of Fluid Mechanics* 643, pp. 233–266.
- Huang, Peng-Qi et al. (2019). "Global Distribution of the Oceanic Bottom Mixed Layer Thickness". In: *Geophysical Research Letters* 46.3, pp. 1547–1554.
- Kundu, Pijush K, Ira M Cohen, and David R Dowling (2015). *Fluid Mechanics*. 6th ed. Elsevier.
- Taylor, John R and Sutanu Sarkar (2008). "Stratification effects in a bottom Ekman layer". In: *Journal of Physical Oceanography* 38.11, pp. 2535–2555.
- Weatherly, Georges L and Paul J Martin (1978). "On the structure and dynamics of the oceanic bottom boundary layer". In: *Journal of Physical Oceanography* 8.4, pp. 557–570.